

Vector Spaces

A *vector space* V is a non-empty set together with a binary function $+$ and unary functions c for every $c \in \mathbb{R}$ satisfying the following axioms:

- 1 $+: V \times V \rightarrow V$, that is, if $u, v \in V$ then $u + v \in V$.
- 2 $+$ is associative.
- 3 $+$ is commutative.
- 4 there is $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
- 5 for every $v \in V$, there is $-v \in V$ such that $v + (-v) = 0$.
- 6 for every $c \in \mathbb{R}$ and $v \in V$, $cv \in V$.
- 7 for every $c \in \mathbb{R}$ and $u, v \in V$, $c(u + v) = cu + cv$.
- 8 for every $c, d \in \mathbb{R}$ and every $u \in V$, $(c + d)u = cu + du$.
- 9 for every $c, d \in \mathbb{R}$ and every $u \in V$, $c(dv) = (cd)v$.
- 10 for every $v \in V$, $1v = v$.

Ignore scalar mult. and just concentrate on a single binary function which we call $+$:

Ex: $\{1, 2, 3, 4, \dots\}$ and we can add these using $+$. Axioms 1, 2, 3 are satisfied

Axiom 4: There is no element⁰ of this set satisfying $x + 0 = x$ for all x . $\times$.

Ex: $\{0, 1, 2, 3, \dots\}$ together with $+$

Axioms 1-3 still hold

Axiom 4: 0 is 0 !

Axiom 5: Is it true that for any x
in our set, we can find $-x$ in our set
s.t. $x + (-x) = 0$.

Ex: $\mathbb{Z}$ with $+$ (integers with $+$).

Axioms 1-5 are good.

Ex: $V = \mathbb{R}_+ = \{r \in \mathbb{R} : r \geq 0\}$

We define $x+y$ in V as xy .

This addition is well-defined.

Mult. is associative and commutative.

Axiom 4: What is 0? 1!

$$x+1 = x \cdot 1 = x \text{ for all } x.$$

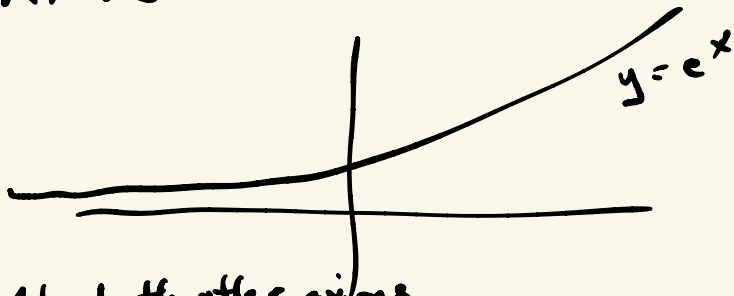
Axiom 5: What is $-x$?

$$-x \text{ is } \frac{1}{x} \quad x + (-x) = x \cdot \frac{1}{x} = 1.$$

Continuing this example, define scalar multiplication by $c \in \mathbb{R}$ as

$$x^c \text{ for all } x \in \mathbb{R}_+ \text{ (Correction)}$$

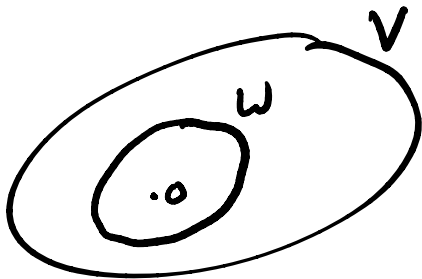
Axiom 6: These functions are well-defined



Check the other axioms.

Subspaces

If V is a vector space and W is a non-empty subset of V then we call W a subspace of V if W is closed under $+$ and scalar multiplication i.e. if $u, v \in W$ then $u + v \in W$ and if $c \in \mathbb{R}$ and $v \in W$ then $cv \in W$.



Fact: If W is a subspace of vector space V then W is also a vector space.

Ex: P = the set of all polynomial functions on $[0,1]$. with the usual + and scalar mults on functions.

Continuous functions on $[0,1]$ is a vector space. Polynomials are continuous.

Need to show P is closed under + and scalar mult.

$$\begin{array}{r} a_0 + a_1x + \dots + a_nx^n \\ b_0 + \dots \qquad \qquad \qquad b_mx^m \\ \hline (a_0+b_0) + (a_1+b_1)x + \dots (a_n+b_n)x^n + \dots b_mx^m \end{array}$$

Conclusion: P is a subspace of the vector space of cont. functions on $[0,1]$ and so P is a vector space.

Ex: $P_n =$ all poly. of degree $\leq n$ on $[0,1]$.

$P_n \subseteq P$ and is closed under + and scalar mult. so P_n is a subspace of P and a vector space.

$P_0 \subseteq P_1 \subseteq P_2 \subseteq \dots \subseteq P_n \subseteq \dots \subseteq P \subseteq \text{Cont.}$
 $\downarrow$
 $\mathbb{R}$

Ex's on
Co.13.

Examples of subspaces

Given an $m \times n$ matrix A , the set of solutions of $Ax = 0$ is a subspace of $\mathbb{R}^n$ called the *nullspace* of A , written $\text{Nul}(A)$.

$$\text{Nul}(A) = \{x \in \mathbb{R}^n : Ax = 0\}$$

Claim: $\text{Nul}(A)$ is a subspace of $\mathbb{R}^n$.

1) $0 \in \text{Nul}(A)$ - $A0 = 0$.

2) If $x, y \in \text{Nul}(A)$, is $x+y \in \text{Nul}(A)$?

$$\begin{aligned} A(x+y) &= Ax + Ay \\ &= 0 + 0 = 0. \end{aligned}$$

3) If $x \in \text{Nul}(A)$ and $c \in \mathbb{R}$, is $cx \in \text{Nul}(A)$?

$$\begin{aligned} A(cx) &= cAx \\ &= c0 = 0 \quad \text{so } cx \in \text{Nul}(A). \end{aligned}$$

So $\text{Nul}(A)$ is a subspace of $\mathbb{R}^n$.

The span as a subspace

Suppose that $v_1, \dots, v_k$ are vectors in $\mathbb{R}^n$. The span of $v_1, \dots, v_k$ is a subspace of $\mathbb{R}^n$. Recall that the span of $v_1, \dots, v_k$ is the set of all linear combinations of $v_1, \dots, v_k$.

Claim: If $v_1, \dots, v_k \in \mathbb{R}^n$ then the span of $v_1, \dots, v_k$ is a subspace of $\mathbb{R}^n$.

- 1) $0v_1 + \dots + 0v_k = 0 \in \text{Span}(v_1, \dots, v_k)$.
- 2) Why is the span closed under $+$?

$$\left. \begin{array}{l} c_1 v_1 + \dots + c_n v_n \\ + \quad d_1 v_1 + \dots + d_n v_n \end{array} \right\} \in \text{Span of } v_1, \dots, v_n.$$

$$(c_1 + d_1) v_1 + \dots + (c_n + d_n) v_n \leftarrow \text{is in the span of } v_1, \dots, v_n.$$

3) The span is closed under scalar mult.

Recall that if A is $m \times n$ then the column space of A , $\text{Col}(A)$, is the span of the columns of A which by the previous claim is a subspace of $\mathbb{R}^m$.